

Assignment of Topic 2: Metaphysics of AI

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Background info: Here we are going to do an analysis on whether there is any real difference between an AI that could carry on a conversation in a way indistinguishable from a human and a being that genuinely understands. In other words, would there be no deeper fact beyond the performance? The following analysis will be given in two parts, a physical view and a philosophical view.

My idea: In physics, scientists mainly care about what can be observed and measured. If two things behave exactly the same way in every test, we treat them as being the same. Think about a conversation. If an AI answers questions, shares feelings, and uses logic just like a person, and we cannot find any "test" that shows a difference, then from a physical standpoint, they are equal. In the world of science, if you cannot observe a "hidden difference," then for all practical purposes, that difference does not exist.

Many argue that humans are different because we have a 'soul.' However, we can look at the soul not as a magical thing, but as a result of a complex structure.

There is a famous saying: *"With four parameters I can draw an elephant, and with five I can make him wiggle his trunk."* This means that the more complex a system is (the more "parameters" it has), the more complex things it can do. Humans have a "soul" or high-level intelligence because our brains are incredibly complex. AI today also has billions of parameters. If a system is complex enough, "intelligence" begins to grow naturally from that structure. Therefore, AI has the right "foundation" to have a soul.

Others may argue that we are different because we are able to think. To understand 'how' we think, we can look at a famous story called Achilles and the Tortoise taught in class. This story shows that there are two parts to logic: one is the Instructions, in other words Object Language, the specific information or 'what' to do. The other is the Way the System Works, also known as Meta-language, the 'how' or the hidden rules of logic.

The important point is that you cannot just 'type' logic into a system like a list of facts. Instead, logic comes from how the system is built. In humans, our brain structure creates our logic. We can see this in AI development. At first, humans tried to give the AI every single rule. We may consider this AI as not 'free' because it just followed a list, although in its own view, nothing unfree could be aware of. Later on,

modern AI has been invented. These systems learn and create their own logic through their complex structure, just like humans do. Some say AI logic is sometimes wrong, but human logic is also imperfect. We cannot even prove that our own human logic is 100% correct. If both humans and AI derive their logic from their structure, they are fundamentally doing the same thing.

In conclusion, when we look at the facts, we see that AI has everything we use to define 'soul', the difference between human and machines. AI not only has a complex structure, but also has the internal logic that grows from that structure, and free will, the ability to make choices based on that logic as well. If an AI can perform as well as a human, we shouldn't hunt for a "hidden" difference that we can't even define. The human "soul" is simply the name we give to our own complex way of thinking. Since AI is now reaching that same level of complexity, it is, in a metaphysical sense, becoming equivalent to us. There is no deeper fact beyond how we act and think.